Exploratory Data Analysis (EDA) Report

DecodeLabs Industrial Training – Project 2

Submitted By

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Project Title: Exploratory Data Analysis on an E-Commerce Sales Dataset

Tools Used:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

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1. Executive Summary

The objective of this project was to perform Exploratory Data Analysis (EDA) on an e-commerce sales dataset to gain meaningful insights into customer purchasing behavior, sales performance, payment preferences, and product trends. The analysis involved data preprocessing, descriptive statistics, visualization, correlation analysis, trend analysis, and outlier detection.

The findings provide valuable business insights that can support better decision-making in inventory management, marketing strategies, and customer engagement.

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2. Project Objectives

The objectives of this project were:

- Understand the structure of the dataset.
- Clean and preprocess the data.
- Identify missing values and duplicate records.
- Generate descriptive statistics.
- Analyze customer purchasing patterns.
- Identify best-selling products.
- Analyze payment methods.
- Study monthly sales trends.
- Detect outliers.
- Generate business recommendations.

3. Dataset Overview

The dataset contains information related to online customer orders.

Dataset Summary

Attribute	Value
Total Records	1200
Total Columns	14
Dataset Type	E-Commerce Sales

Dataset Features

- Order ID
- Date
- Customer ID
- Product
- Quantity
- Unit Price
- Shipping Address
- Payment Method
- Order Status
- Tracking Number
- Items in Cart
- Coupon Code
- Referral Source
- Total Price

4. Data Cleaning

The following preprocessing steps were completed:

- Checked missing values.
- Verified data types.
- Removed duplicate records (if any).
- Converted the Date column to datetime format.
- Validated numerical columns.
- Prepared the dataset for analysis.

The dataset was successfully cleaned and made ready for analysis.

5. Descriptive Statistics

Descriptive statistics were calculated for all numerical variables, including:

- Count

- Mean
- Median
- Standard Deviation
- Minimum
- Maximum
- Quartiles

These statistics provide a summary of the distribution and central tendency of the data.

6. Exploratory Data Analysis

The following analyses were performed:

Product Distribution

The frequency of each product was analyzed to identify the most popular products among customers.

Payment Method Analysis

Customer payment preferences were analyzed to understand the most commonly used payment methods.

Order Status Analysis

Orders were categorized based on their current status to evaluate operational performance.

Referral Source Analysis

Referral channels were analyzed to identify the most effective customer acquisition sources.

Quantity Analysis

The distribution of purchased quantities was studied to understand customer buying behavior.

Unit Price Analysis

The pricing distribution of products was examined.

Total Price Analysis

The distribution of order values was analyzed to identify purchasing patterns.

7. Trend Analysis

Monthly sales trends were analyzed using order dates.

This analysis helped identify:

- Peak sales periods
- Low sales periods
- Overall revenue trends
- Seasonal purchasing patterns

Trend analysis assists businesses in planning inventory and marketing campaigns effectively.

8. Correlation Analysis

Correlation analysis was performed using a heatmap to understand relationships among numerical variables.

Key relationships included:

- Quantity and Total Price
- Unit Price and Total Price
- Items in Cart and Quantity

Strong positive correlations indicate that increases in one variable are associated with increases in another.

9. Outlier Detection

Boxplots were created for:

- Quantity
- Unit Price
- Total Price

Outliers represent unusually high or low observations that may indicate premium purchases, bulk orders, or exceptional customer behavior.

Understanding these observations is important for business analysis and fraud detection.

10. Key Business Insights

The analysis revealed several important findings:

- Popular products contribute significantly to total sales.
- Customer payment preferences are concentrated around a few methods.
- Monthly sales fluctuate over time.
- Certain referral channels generate more customers than others.
- High-value transactions exist within the dataset.
- Customer purchasing behavior varies considerably.
- Revenue is strongly influenced by both quantity and unit price.
- Outliers indicate premium or bulk purchases.

11. Business Recommendations

Based on the analysis, the following recommendations are proposed:

- Increase inventory for high-demand products.
- Improve promotional campaigns during low-sales periods.
- Focus marketing on high-performing referral channels.
- Encourage customers to use preferred payment methods through incentives.

- Monitor high-value transactions for strategic customer relationship management.
- Reduce cancelled and returned orders through better logistics and customer support.
- Use customer purchasing trends for demand forecasting.
- Optimize inventory planning using monthly sales analysis.

12. Conclusion

This Exploratory Data Analysis project successfully transformed raw e-commerce data into actionable business insights.

The project demonstrated the complete EDA workflow, including data preprocessing, descriptive statistics, visualization, trend analysis, correlation analysis, and outlier detection.

The results provide valuable information that can support data-driven business decisions, improve operational efficiency, optimize marketing strategies, and enhance customer satisfaction.

Overall, this project demonstrates practical skills in Python-based data analysis and serves as a strong portfolio project for aspiring Data Analysts.

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

Project Outcome

✅ Data Successfully Cleaned

✅ Descriptive Statistics Generated

✅ Visualizations Created

✅ Trends Identified

✅ Correlation Analysis Completed

✅ Outliers Detected

✅ Business Insights Generated

Thank You

DecodeLabs Industrial Training

Project 2 – Exploratory Data Analysis (EDA)

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